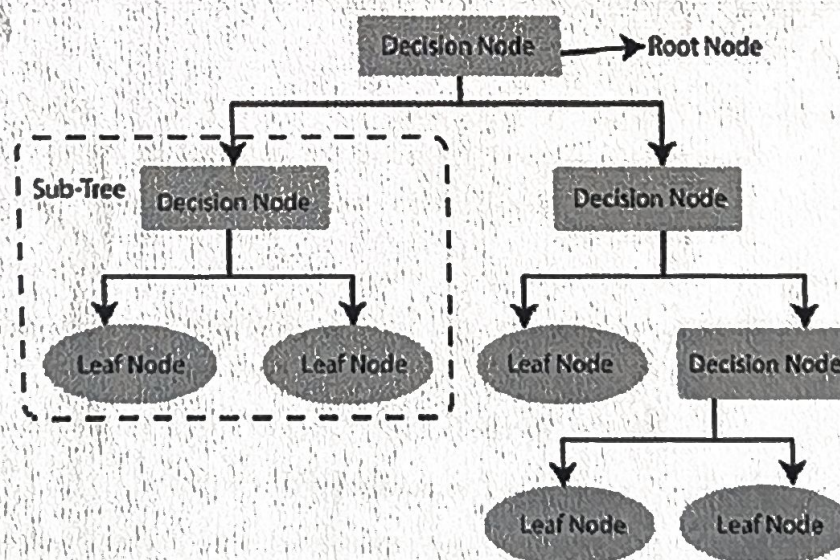


Decision Tree Classification Algorithm

- Decision Tree is a Supervised learning technique that can be used for both classification and Regression problems, but mostly it is preferred for solving Classification problems. It is a tree-structured classifier, where **internal nodes represent the features of a dataset, branches represent the decision rules and each leaf node represents the outcome.**
- In a Decision tree, there are two nodes, which are the Decision Node and Leaf Node. Decision nodes are used to make any decision and have multiple branches, whereas Leaf nodes are the output of those decisions and do not contain any further branches.
- The decisions or the test are performed on the basis of features of the given dataset.
- *It is a graphical representation for getting all the possible solutions to a problem/decision based on given conditions.*
- It is called a decision tree because, similar to a tree, it starts with the root node, which expands on further branches and constructs a tree-like structure.
- In order to build a tree, we use the CART algorithm, which stands for **Classification and Regression Tree** algorithm.
- A decision tree simply asks a question, and based on the answer (Yes/No), it further split the tree into subtrees.
- Below diagram explains the general structure of a decision tree:



Why use Decision Trees?

There are various algorithms in Machine learning, so choosing the best algorithm for the given dataset and problem is the main point to remember while creating a machine learning model. Below are the two reasons for using the Decision tree:

- Decision Trees usually mimic human thinking ability while making a decision, so it is easy to understand.
- The logic behind the decision tree can be easily understood because it shows a tree-like structure.

Decision Tree Terminologies

- **Root Node:** Root node is from where the decision tree starts. It represents the entire dataset, which further gets divided into two or more homogeneous sets.
- **Leaf Node:** Leaf nodes are the final output node, and the tree cannot be segregated further after getting a leaf node.
- **Splitting:** Splitting is the process of dividing the decision node/root node into sub-nodes according to the given conditions.
- **Branch/Sub Tree:** A tree formed by splitting the tree.
- **Pruning:** Pruning is the process of removing the unwanted branches from the tree.
- **Parent/Child node:** The root node of the tree is called the parent node, and other nodes are called the child nodes.

How does the Decision Tree algorithm Work?

In a decision tree, for predicting the class of the given dataset, the algorithm starts from the root node of the tree. This algorithm compares the values of root attribute with the record (real dataset) attribute and, based on the comparison, follows the branch and jumps to the next node.

For the next node, the algorithm again compares the attribute value with the other sub-nodes and move further. It continues the process until it reaches the leaf node of the tree. The complete process can be better understood using the below algorithm:

- **Step-1:** Begin the tree with the root node, says S, which contains the complete dataset.
- **Step-2:** Find the best attribute in the dataset using **Attribute Selection Measure (ASM)**.
- **Step-3:** Divide the S into subsets that contains possible values for the best attributes.
- **Step-4:** Generate the decision tree node, which contains the best attribute.
- **Step-5:** Recursively make new decision trees using the subsets of the dataset created in step -3. Continue this process until a stage is reached where you cannot further classify the nodes and called the final node as a leaf node.

Attribute Selection Measures

While implementing a Decision tree, the main issue arises that how to select the best attribute for the root node and for sub-nodes. So, to solve such problems there is a technique which is called as **Attribute selection measure or ASM**. By this measurement, we can easily select the best attribute for the nodes of the tree. There are two popular techniques for ASM, which are:

- **Information Gain**
- **Gini Index**

1. Information Gain:

- Information gain is the measurement of changes in entropy after the segmentation of a dataset based on an attribute.
- It calculates how much information a feature provides us about a class.
- According to the value of information gain, we split the node and build the decision tree.
- A decision tree algorithm always tries to maximize the value of information gain, and a node/attribute having the highest information gain is split first. It can be calculated using the below formula:

Entropy: Entropy is a metric to measure the impurity in a given attribute. It specifies randomness in data. Entropy can be calculated as:

$$\text{Entropy}(s) = -P(\text{yes}) \log_2 P(\text{yes}) - P(\text{no}) \log_2 P(\text{no})$$

Where,

- **S = Total number of samples**
- **P(yes) = probability of yes**
- **P(no) = probability of no**

Pruning: Getting an Optimal Decision tree

Pruning is a process of deleting the unnecessary nodes from a tree in order to get the optimal decision tree.

A too-large tree increases the risk of overfitting, and a small tree may not capture all the important features of the dataset. Therefore, a technique that decreases the size of the learning tree without reducing accuracy is known as Pruning. There are mainly two types of tree pruning technology used:

- **Cost Complexity Pruning**
- **Reduced Error Pruning.**

Decision Tree in Machine Learning

1. Basic Structure of a Decision Tree

A Decision Tree is a supervised machine learning algorithm used for both classification and regression problems. It makes decisions by repeatedly splitting the dataset based on feature values, forming a tree-like structure.

A decision tree consists of:

1. Root Node – The first and most important decision/splitting feature.
2. Internal/Decision Nodes – Conditions used to divide the data.
3. Branches – Outcomes of the decision.
4. Leaf/Terminal Nodes – Final predicted class or value.

2. How Does a Decision Tree Work?

Step 1: Start with the complete dataset.

All training observations are initially placed at the root node.

Step 2: Select the best feature.

The algorithm evaluates different features and chooses the one that produces the best separation of classes.

Step 3: Split the dataset.

The selected feature divides the dataset into smaller subsets.

Step 4: Repeat recursively.

The same procedure is applied to each subset.

Step 5: Stop splitting.

Splitting stops when all observations belong to the same class, no useful feature remains, maximum tree depth is reached, minimum number of samples is reached, or further splitting does not significantly improve the model.

3. Entropy

Entropy measures the impurity or uncertainty of a dataset.

$$\text{Entropy}(S) = - \sum p_i \log_2(p_i)$$

where:

S = dataset

c = number of classes

p_i = proportion of observations belonging to class i

For two classes:

$$\text{Entropy}(S) = -p_1 \log_2(p_1) - p_2 \log_2(p_2)$$

Interpretation:

- All observations belong to one class $\rightarrow$ Entropy = 0
- Equal observations in two classes $\rightarrow$ Entropy = 1
- Mixed classes $\rightarrow$ Entropy is between 0 and 1

Thus, lower entropy means a purer dataset.

4. Information Gain

Information Gain measures how much uncertainty is reduced after splitting the data.

$$\text{IG}(S,A) = \text{Entropy}(S) - \sum [|S_v|/|S| \times \text{Entropy}(S_v)]$$

where:

A = selected attribute

S_v = subset corresponding to value v

The feature having the highest Information Gain is generally selected for splitting.

5. Advantages of Decision Trees

1. Easy to understand and interpret.
2. Can handle both numerical and categorical variables.
3. Requires relatively little data preprocessing.
4. Does not require feature scaling.
5. Can model nonlinear relationships.
6. Useful for both classification and regression.
7. Easy to visualize.
8. Feature importance can be calculated.

6. Disadvantages

1. Can easily overfit the training data.
2. Small changes in data can produce a very different tree.
3. Deep trees can become complicated.
4. A single decision tree may have lower predictive performance than ensemble methods.
5. It can be biased toward features with many possible split points/categories.
6. Greedy splitting does not necessarily produce the globally optimal tree.

7. How to Prevent Overfitting

Important DecisionTreeClassifier parameters include:

- max_depth
- min_samples_split
- min_samples_leaf
- max_leaf_nodes
- criterion
- ccp_alpha

Pre-pruning:

Stop the tree from growing too much, for example by limiting max_depth or increasing min_samples_leaf.

Post-pruning:

First grow the tree and then remove branches that do not contribute sufficiently to predictive performance.

8. Decision Tree Algorithm – Summary

Start

↓

Collect Training Dataset

↓

Calculate Impurity

↓

Evaluate Possible Splits

↓

Calculate Information Gain / Gini

↓

Select Best Feature

↓

Split Dataset

↓

Create Child Nodes

↓

Is stopping condition satisfied?

├─ No → Continue splitting

└─ Yes → Create Leaf Node

↓

Prediction

In one line:

Dataset → Best Split → Child Nodes → Recursive Splitting → Leaf/Prediction

Decision Tree

Ex

Day	Weather	Temperature	Humidity	Wind	Play Football
Day 1	Sunny	Hot	High	Weak	No
Day 2	Sunny	Hot	High	Strong	No
Day 3	Cloudy	Hot	High	Weak	Yes
Day 4	Rain	Mild	High	Weak	Yes
Day 5	Rain	Cool	Normal	Weak	Yes
Day 6	Rain	Cool	Normal	Strong	No
Day 7	Cloudy	Cool	Normal	Strong	Yes
Day 8	Sunny	Mild	High	Weak	No
Day 9	Sunny	Cool	Normal	Weak	Yes
Day 10	Rain Sunny	Cool Mild	Normal	Weak	Yes
Day 11	Sunny	Mild	Normal	Strong	Yes
Day 12	Cloudy	Mild	High	Strong	Yes
Day 13	Cloudy	Hot	Normal	Weak	Yes
Day 14	Rain	Mild	High	Strong	No

1. First find root node:

2. DT based on two

① Entropy

② Information gain

3. Calculate IG (Information gain for all features)
highest IG of feature will be root node.

Now: IG for all feature.

Step 1: Calculate IG for Weather

Step 1: Entropy of entire dataset.

$$S(+9, -5) = -\frac{9}{14} \log_2 \frac{9}{14} - \frac{5}{14} \log_2 \frac{5}{14} = 0.94$$

Step 2: Entropy for all attributes (in weather feature)

weather $\begin{cases} \rightarrow \text{Cloudy} \\ \rightarrow \text{Rain} \\ \rightarrow \text{Sunny} \end{cases}$

$$\text{Entropy of Sunny } \{+2, -3\} = -\frac{2}{5} \log_2 \frac{2}{5} - \frac{3}{5} \log_2 \frac{3}{5} = 0.92$$

$$\text{Entropy of Cloudy } \{+4, 0\} = -\frac{4}{4} \log_2 \frac{4}{4} - \frac{0}{4} \log_2 \frac{0}{4} = 0$$

$$\text{Entropy of Rain } \{+3, -2\} = -\frac{3}{5} \log_2 \frac{3}{5} - \frac{2}{5} \log_2 \frac{2}{5} = 0.92$$

$$\text{Information gain} = \text{Entropy (whole data)} - \frac{5}{14} \text{Ent}(S) - \frac{4}{14} \text{Ent}(C) - \frac{5}{14} \text{Ent}(R) \\ = 0.246$$

2. Calculate IG of Temperature

Step 1: Entropy of entire dataset

$$S\{+9, -5\} = -\frac{9}{14} \log_2 \frac{9}{14} - \frac{5}{14} \log_2 \frac{5}{14} = 0.94$$

Step 2: Entropy of all attributes

$$\text{Entropy of Hot } \{+2, -2\} = -\frac{2}{4} \log_2 \frac{2}{4} - \frac{2}{4} \log_2 \frac{2}{4} = 1$$

$$\text{Entropy of Mild } \{+4, -2\} = -\frac{4}{6} \log_2 \frac{2}{6} - \frac{2}{6} \log_2 \frac{2}{6} = 0.91$$

$$\text{Entropy of Cold } \{+3, -1\} = -\frac{3}{4} \log_2 \frac{3}{4} - \frac{1}{4} \log_2 \frac{1}{4} = 0.81$$

$$\begin{aligned} \text{Information gain} &= \text{Ent}(\text{whole data}) - \frac{4}{14} \text{Ent}(H) - \frac{6}{14} \text{Ent}(M) \\ &\quad - \frac{4}{14} \text{Ent}(C) \\ &= 0.029 \end{aligned}$$

3. Calculate IG of Humidity

Step 1: Entropy of entire dataset

$$S\{+9, -5\} = -\frac{9}{14} \log_2 \frac{9}{14} - \frac{5}{14} \log_2 \frac{5}{14}$$

Step 2: Entropy of all attributes:

$$\text{Entropy of High } \{+3, -1\} = -\frac{3}{7} \log_2 \frac{3}{7} - \frac{1}{7} \log_2 \frac{1}{7} = 0.98$$

$$\text{Entropy of Normal } \{+6, -1\} = -\frac{6}{7} \log_2 \frac{6}{7} - \frac{1}{7} \log_2 \frac{1}{7} = 0.59 \quad (R)$$

$$\begin{aligned} \text{Information gain} &= \text{Entropy}(\text{whole data}) - \frac{7}{14} \text{Ent}(H) - \frac{7}{14} \text{Ent}(N) \\ &= 0.15 \end{aligned}$$

1. Calculate IG of Wind.

Step 1: Entropy of entire dataset

$$S \{+9, -5\} = -\frac{9}{14} \log_2 \frac{9}{14} - \frac{5}{14} \log_2 \frac{5}{14} = 0.94$$

Step 2: Entropy of all attributes:

$$\text{Entropy of strong } \{+3, -3\} = -\frac{3}{6} \log_2 \frac{3}{6} - \frac{3}{6} \log_2 \frac{3}{6} = 1.0$$

$$\text{Entropy of Normal } \{+6, -2\} = -\frac{6}{8} \log_2 \frac{6}{8} - \frac{2}{8} \log_2 \frac{2}{8} = 0.81$$

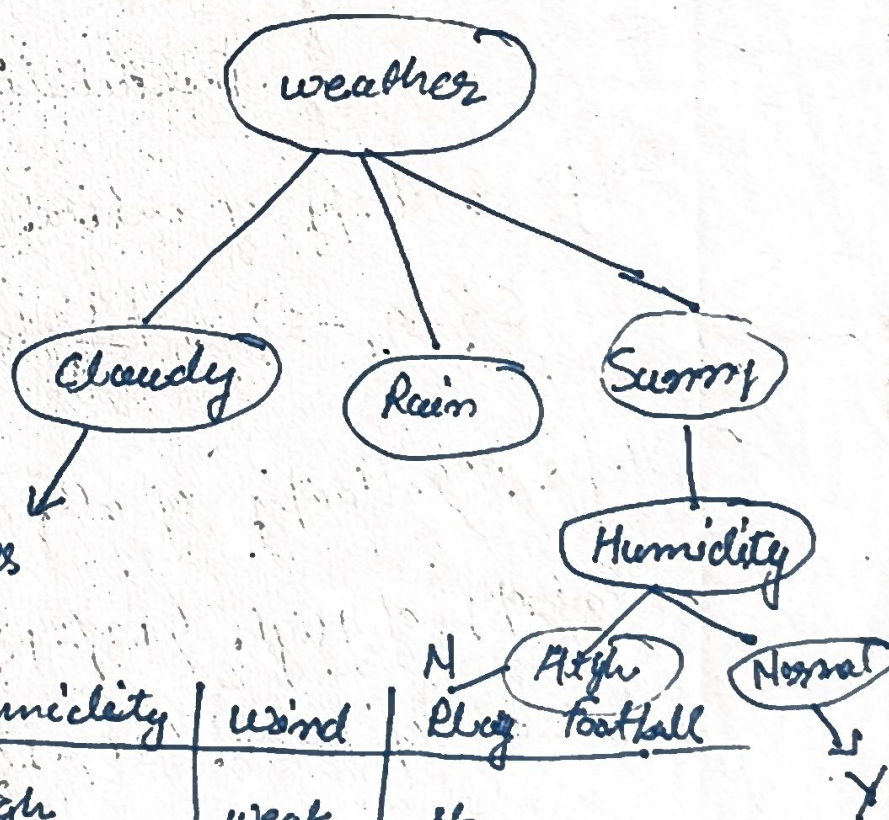
$$\begin{aligned} \text{Information gain of wind} &= \text{Ent(whole data)} - \frac{6}{14} \text{Ent(S)} - \frac{8}{14} \text{Ent(N)} \\ &= 0.94 - 0.0478 \end{aligned}$$

1. Gain (S, weather) = 0.246 ✓

2. Gain (S, Temp) = 0.029

3. Gain (S, Humidity) = 0.15

4. Gain (S, wind) = 0.0478



Now take Sunny:

Day	Weather	Temperature	Humidity	Wind	N
Day 1	Sunny	Hot	High	weak	No
Day 2	Sunny	Hot	High	Strong	No
Day 3	Sunny	Mild	High	Weak	No
Day 4	Sunny	Cool	Normal	weak	Yes
Day 5	Sunny	Mild	Normal	Strong	Yes

Calculate IG of Temperature

Step 1: Entropy of sunny $\{+2, -3\} = -\frac{2}{5} \log_2 \frac{2}{5} - \frac{3}{5} \log_2 \frac{3}{5} = 0.97$

Step 2 Entropy of all attributes:

Entropy of Hot $\{+0, -2\} = -\frac{0}{2} \log_2 \frac{0}{2} - \frac{2}{2} \log_2 \frac{2}{2} = 0$

Entropy of Mild $\{+1, -1\} = -\frac{1}{2} \log_2 \frac{1}{2} - \frac{1}{2} \log_2 \frac{1}{2} = 1.0$

Entropy of Cool $\{+1, -0\} = -\frac{1}{1} \log_2 \frac{1}{1} - \frac{0}{1} \log_2 \frac{0}{1} = 0$

Information gain = Entropy(Sunny) = $-\frac{2}{5} \text{Ent}(H) - \frac{2}{5} \text{Ent}(M) - \frac{1}{5} \text{Ent}(C)$
 $= 0.57$

Calculate IG of Humidity

Step 1: Entropy of sunny $\{+2, -3\} = -\frac{2}{5} \log_2 \frac{2}{5} - \frac{3}{5} \log_2 \frac{3}{5} = 0.97$

Step 2: Entropy of all attributes:

Entropy of Hot high $\{+0, +3\} = -\frac{0}{3} \log_2 \frac{0}{3} - \frac{3}{3} \log_2 \frac{3}{3} = 0$

Entropy of Normal $\{+2, 0\} = -\frac{2}{2} \log_2 \frac{2}{2} - \frac{0}{2} \log_2 \frac{0}{2} = 0$

Information gain = Ent(Sunny) = $-\frac{3}{5} \text{Ent}(h) - \frac{2}{5} \text{Ent}(N)$
 $= 0.97$

Calculate IG of humidity: wind

Step 1: Entropy of wind $\{+2, -3\} = -\frac{2}{5} \log_2 \frac{2}{5} - \frac{3}{5} \log_2 \frac{3}{5} = 0.97$

Step 2: Entropy of all attributes:

$$\text{Entropy of weak } \{+1, -2\} = -\frac{1}{3} \log_2 \frac{1}{3} - \frac{2}{3} \log_2 \frac{2}{3} = 0.918$$

$$\text{Entropy of strong } \{+1, -1\} = -\frac{1}{2} \log_2 \frac{1}{2} - \frac{1}{2} \log_2 \frac{1}{2} = 1$$

$$\begin{aligned} \text{Information gain} &= \text{Ent}(\text{wind}) - \frac{3}{5} \text{Ent}(w) - \frac{2}{5} \text{Ent}(s) \\ &= 0.97 - \frac{3}{5} \times 0.918 - \frac{2}{5} \times 1 \\ &= 0.97 - 0.5508 - 0.4 \\ &= 0.97 - 0.9508 \\ &= 0.0192 \end{aligned}$$

$$\text{Gain} (S_{\text{sunny}}, \text{Temp}) = 0.57$$

$$\text{Gain} (S_{\text{sunny}}, \text{humidity}) = 0.97$$

$$\text{Gain} (S_{\text{sunny}}, \text{wind}) = 0.019$$

Day	Weather	Temperature	Humidity	wind	Play Football
Day 4	Rain	Mild	High	Weak	yes
Day 5	Rain	Cool	Normal	Weak	yes
Day 6	Rain	Cool	Normal	Strong	No
Day 10	Mild Rain	Mild Normal	Normal Weak	Weak	Yes
Day	Rain	Mild	High	Strong	No